

Karthik Thyagarajan

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EDUCATION

Purdue University

West Lafayette, Indiana

B.S. of Computer Science & Artificial Intelligence (Double Major), GPA: 3.93

May 2027

- **Coursework:** Data Structures & Algorithms, Computer Architecture, Programming in C, Competitive Programming, OOP

EXPERIENCE

AI Research Intern

May 2026 – Aug 2026

Samsung Research America

Mountain View, CA

- Built an on-device signal router for always-on ambient AI on Bixby Edge AI, filtering device signals on-phone at 95% suppression and 90% intent-routing accuracy to reserve frontier models for events that need them.
- Co-architected Samsung's always-on cross-device agent orchestrator from scratch, owning half of a 25-module Kotlin runtime that ships on XR glasses and gives phone, glasses, and outside developers one platform to build agents on.
- "Built the runtime's safety and control plane": 'capability-based permissions, user confirmation flows, policy-gated tool dispatch, token and cost budgets, and PII-redacted audit logging, so every action an agent takes is gated, bounded, and recorded.'
- Built phone-control tooling that gives agents computer-use-style access to any app's UI under stale-screen and confirmation guards, plus native calendar, alarm, location, and health tools that reach apps with no exposed API.

Machine Learning Engineering Intern

Jun 2025 – May 2026

Peraton Labs

Silver Spring, MD

- Developed RL agent for IoT malware detection using PyTorch and graph neural networks, reducing exploration latency by 35% and increasing detection coverage by 25% across 500K+ daily device events.
- Built large-scale data analytics ETL pipelines and heterogeneous graph architecture with autoencoders to model device communication patterns, accelerating policy convergence by 40%.

Computer Vision Researcher

Feb 2025 – Aug 2025

Memories.ai

Remote

- Architected video memory framework for AR applications processing 10K+ streams using Python, Flask, and PostgreSQL, achieving 60% throughput improvement through frame sampling optimization.
- Published Python SDK on PyPI with 2K+ downloads, implementing async processing and REST API integration for video analysis workflows.

Undergraduate Robotics Researcher

Mar 2025 – Jun 2025

IDEAS Lab

West Lafayette, IN

- Built real-time SLAM pipeline in C++ and Python with sensor fusion and Kalman filtering, improving 3D reconstruction accuracy by 25% and reducing latency by 30%.
- Implemented neural radiance fields for novel view synthesis, generating photorealistic scene reconstructions for autonomous navigation.

Data Engineering Intern

Aug 2024 – Dec 2024

AgRPA

West Lafayette, IN

- Built end-to-end weed detection pipeline processing 200GB+ drone imagery with Python and cloud storage, implementing parallel ETL workflows and optimized data partitioning for 40% faster analysis.
- Engineered segmentation models achieving 92% accuracy on 50K+ images through agile sprints, reducing herbicide usage by 60% and costs by \$150K annually.

ML Science & Engineering Apprenticeship

Jun 2023 – Aug 2023

Naval Research Laboratory

Washington, D.C.

- Developed ML models improving underwater transmission loss prediction by 20% over physics-based simulations.
- Built LangChain RAG system for classified document retrieval, reducing query time by 65%.

PROJECTS & CLUBS

- Repple** | *Python (FastAPI), Supabase, PostgreSQL, OpenAI API, Pydantic* Ongoing
- Built realtime multiplayer backend with synchronized workout sessions for iOS app with 200+ users.
 - Developed a competitive gamification engine featuring custom ELO matchmaking algorithms, automated cron-based point aggregation, and generative AI workout plans that adapt to user performance metrics.
- buildpurdue** | *Next.js, PostgreSQL, Supabase, UI/UX Design, Leadership, Mentorship* Ongoing
- Co-founded a campus accelerator for 15 student startups, providing resources to validate core problems and refine go-to-market strategies.
 - Forged partnerships with 4 VC firms and industry mentors, allowing student founders to secure feedback and potential pre-seed funding.
- google-tools-mcp** | *Python, MCP, Google APIs, OAuth, Pydantic* Ongoing
- Built MIT-licensed MCP server unifying 169 tools across Drive, Docs, Sheets, Slides, Gmail, Calendar, and Forms behind a single OAuth flow with auto re-auth and dynamic scope handling.
 - Designed read-before-edit guards, lazy authentication, and multi-account profile support to enable safe agent orchestration across the full Google Workspace from one context.
- Caladrius (HackGT - 2nd Place)** | *React Native, Python, LangGraph, AWS S3, REST API, Next.js* Sep 2025
- Architected AI triage system with encrypted, zero-knowledge data transfer for HIPAA compliance.
 - Built multi-agent diagnostic system with LangGraph achieving 85% triage accuracy on symptom analysis.
- Verbatim** | *Next.js, Python, OpenAI API, Google Cloud, Vercel, REST API* Feb 2025
- Created video processing platform integrating Whisper, GPT-4o, Eleven Labs, and Twelve Labs APIs for transcription, translation, voice cloning, and lip-sync, processing 500+ videos.
 - Architected serverless async pipeline on Vercel handling 2GB files with job queues and webhook notifications, reducing wait time by 85%.
- Photonic Implementation of Quantum Key Distribution** | *Python, NumPy, Oscilloscope, Optics Hardware* Oct 2023 – M
- Built polarization-based QKD prototype achieving 95% photon detection rate and demonstrating secure key exchange over 5m fiber optic channel.
 - Automated signal processing pipeline with Python and NumPy for bit extraction and QBER analysis, processing 10K+ samples; published on arXiv.
- Quantum Racer** | *Java, Android SDK, Gradle, XML* Aug 2022 – Dec 2022
- Designed educational Android game simulating quantum mechanics with custom physics engine and MVC architecture using Java.
 - Distributed APK for educational outreach, reaching 100+ students in quantum computing workshops.

TECHNICAL SKILLS

Languages: Python, Java, Kotlin, C/C++, JavaScript, SQL, React, Flask, Next.js, Android
Cloud & DevOps: AWS, S3, EC2, Lambda, GCP, Docker, Git, CI/CD, Linux, REST API
AI/ML: LLM (LangChain, RAG, CoT/Reasoning, RLHF), Agents, MCP, PyTorch, GAN, RL, Diffusion
Quantum Computing: Qiskit, VQE, QAOA, Quantum ML